		$= 0.012$ N
4	**B**	Taking rightward as positive, By relative speed of approach = relative speed of separation $v - 0 = v_Q - \left(-\frac{3v}{5}\right)$ $v_Q = \frac{2v}{5}$ Kinetic energy of Q $= \frac{1}{2}(4m)\left(\frac{2v}{5}\right)^2$ $= \frac{8}{25}mv^2$
5	**C**	Assume that the horizontal force at the top hinge is to the right